figure, though, is that it fails to illustrate how the warps and curves in space evolve when mass and energy move or in some way change their configuration.[3] General relativity predicts that, just as a trampoline assumes a fixed, warped shape if you stand perfectly still, but heaves when you jump up and down, space can assume a fixed, warped shape if matter is perfectly still, as assumed in Figure 3.10, but ripples undulate through its fabric when matter moves to and fro. Einstein came to this realization between 1916 and 1918, when he used the newly fashioned equations of general relativity to show that—much as electric charges racing up and down a broadcast antenna produce electromagnetic waves (this is how radio and television waves are produced)—matter racing this way and that (as in a supernova explosion) produces gravitational waves. And since gravity is curvature, a gravitational wave is a wave of curvature. Just as tossing a pebble into a pond generates outward-spreading water ripples, gyrating matter generates outward-spreading spatial ripples; according to general relativity, a distant supernova explosion is like a cosmic pebble that's been tossed into a spacetime pond, as illustrated in Figure 14.2. The figure highlights an important distinguishing feature of gravitational waves: unlike electromagnetic, sound, and water waves—waves that travel *through* space—gravitational waves travel *within* space. They are traveling distortions in the geometry of space itself.

While gravitational waves are now an accepted prediction of general relativity, for many years the subject was mired in confusion and controversy, at least in part because of overadherence to Machian philosophy. If general relativity fully incorporated Mach's ideas, then the "geometry of space" would merely be a convenient language for expressing the location and motion of one massive object with respect to another. Empty space, in this way of thinking, would be an empty concept, so how could it be sensible to speak of empty space wiggling? Many physicists tried to prove that the supposed waves in space amounted to a misinterpretation of the mathematics of general relativity. But in due course, the theoretical analyses converged on the correct conclusion: gravitational waves are real, and space *can* ripple.